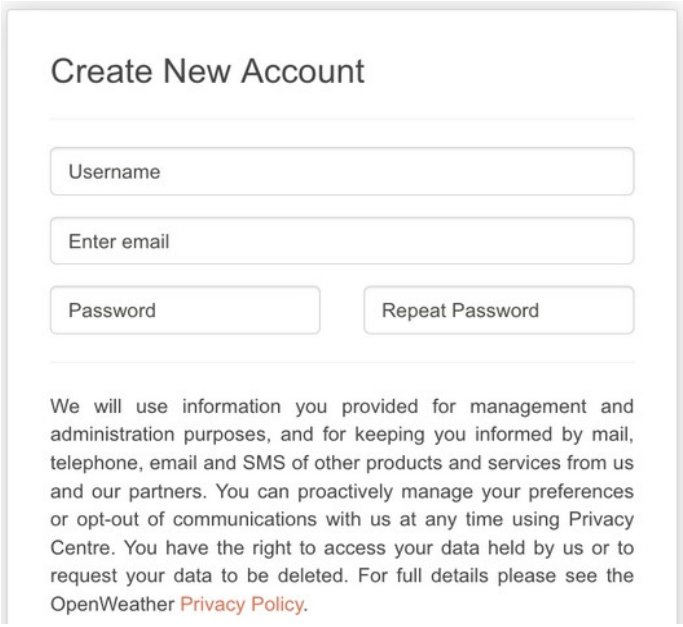
G4Cast Weather Display Project

Software Installation Guide

I strongly recommend installing the software before beginning the assembly process. This ensures that all components are functioning properly and that the device powers up as expected, allowing you to identify and resolve any potential issues early.

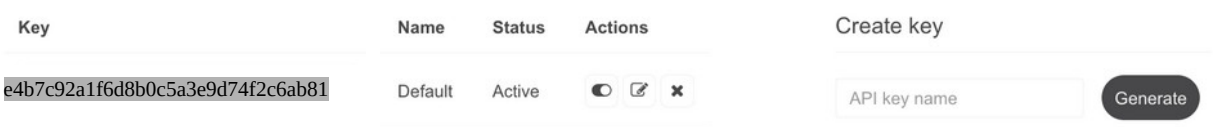
This document serves as the software installation guide for the G4Cast Weather Display project. It provides the essential steps and information required to correctly set up and install the software components and prepare the device for operation.

1. Open https://openweathermap.org/ and create an account using the Sign Up menu.



2. After Creating and account sing in, an go to the API Keys menu https://home.openweathermap.org/api_keys

3. Generate an API key and save it—you will need it later.



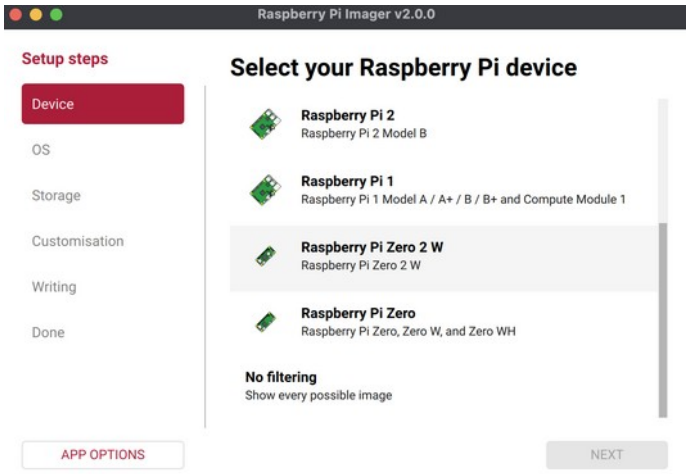
4. Download the **G4Cast.img.zip** image file form google drive to your computer: https://drive.google.com/file/d/1iBaJRY88v7Bh_QprOrG57x_IUyxMpnTC/view

5. Open https://www.raspberrypi.com/software/ and install the Raspberry Pi Imager on your computer.

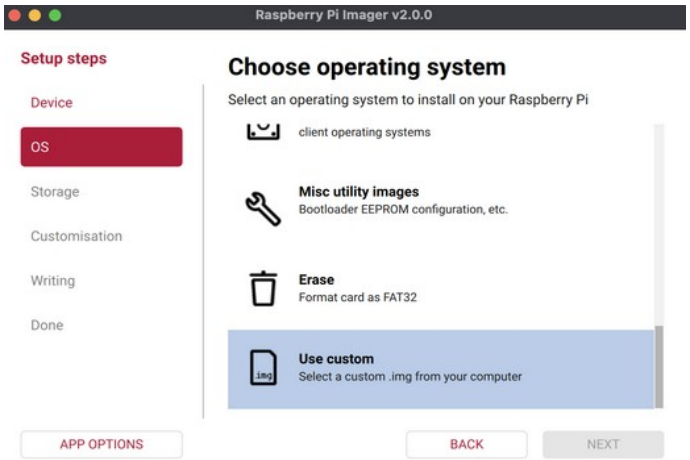


6. Connect your microSD card to your computer.

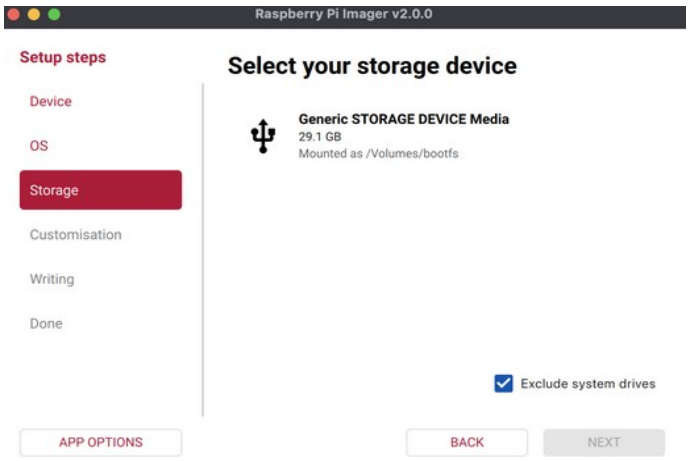
7. Open the Raspberry Pi Imager and select *Raspberry Pi Zero 2 W* from the device list, then click *Next*.



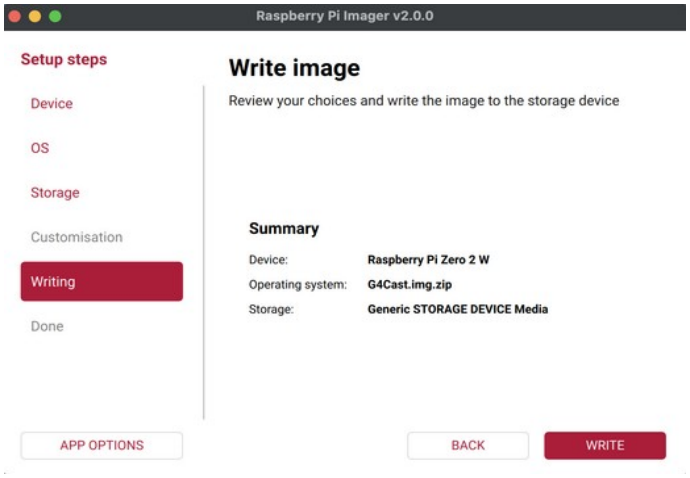
8. In the OS tab, click on "Use custom", then select the previously downloaded **G4Cast.img.zip** file and click Next.



9. In the Storage tab, select your SD card. Be careful: choosing any other option can cause data loss.

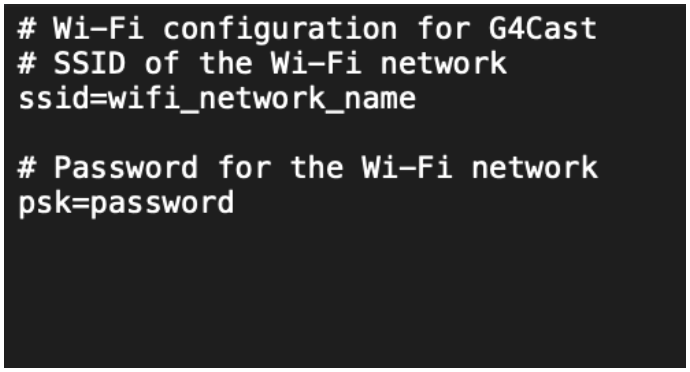


10. Click on Write, and wait until the sc card writing is finished.

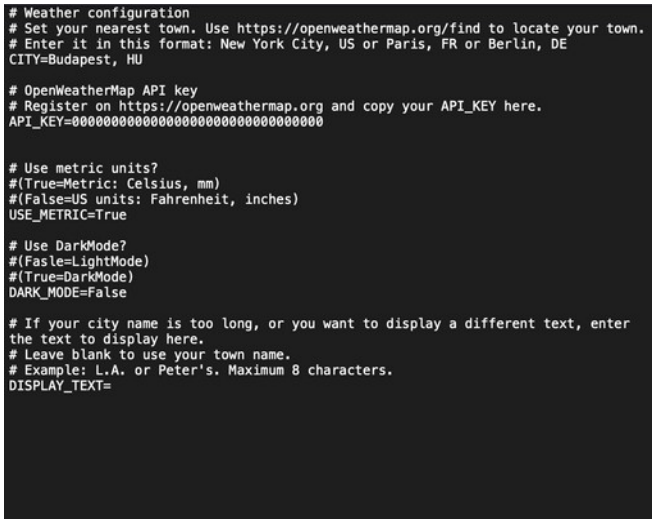


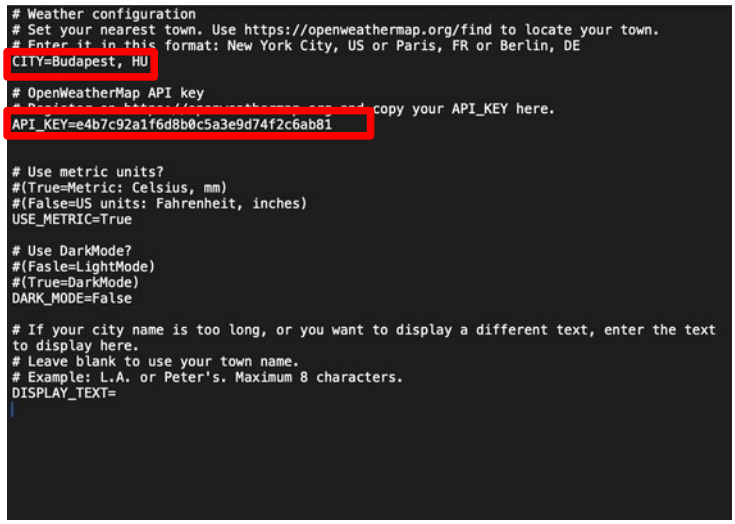
11. Once the writing process is complete, remove the microSD card from your computer and then reconnect it.

12. On the microSD card, open the *G4CAST* folder and then the ***Wifi.txt*** file. Enter your Wi-Fi **network name** and **password**, then save the file.



13. On the microSD card, open the *G4CAST* folder and then the ***G4CastData.txt*** file. Enter your **city** name and your **API key**, then save the file. (You can also adjust the other settings here if you wish.)





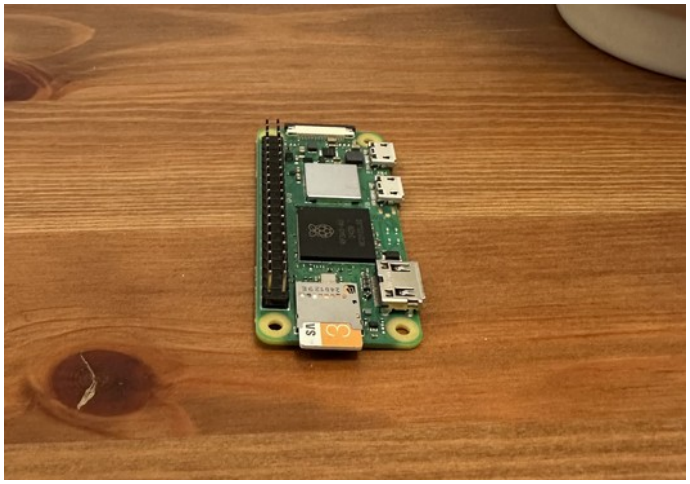
Once you've completed these steps, the software setup is finished. I strongly recommend testing everything now to make sure it works properly before you start assembling the G4 Cast Weather Display.

Follow these steps to check that everything is working correctly:

1. **Prepare the following items:** the preinstalled microSD card you just set up, the E-ink display with HAT and cable, a Raspberry Pi Zero 2 W, and a charger.



2. Insert the preinstalled microSD card into the Raspberry Pi Zero 2 W



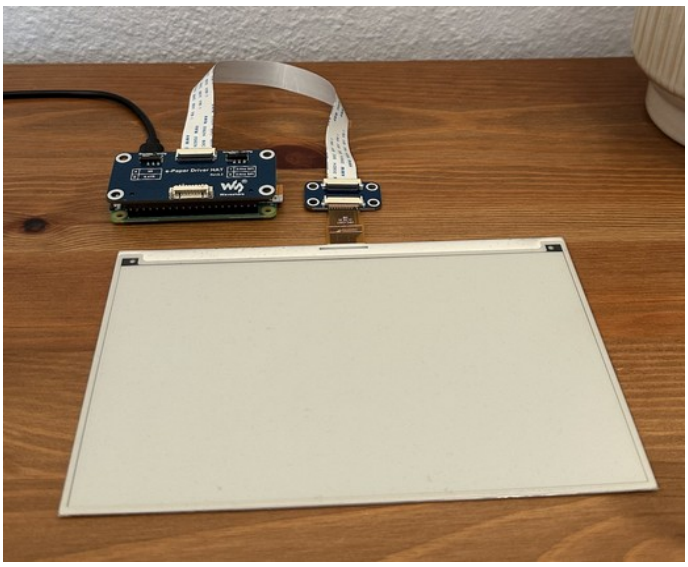
3. Attach the e-ink HAT to the Raspberry Pi Zero 2 W.



4. Attach the E-ink display cable to the Raspberry Pi HAT.



5. Connect the Raspberry Pi to power using the micro-USB cable.

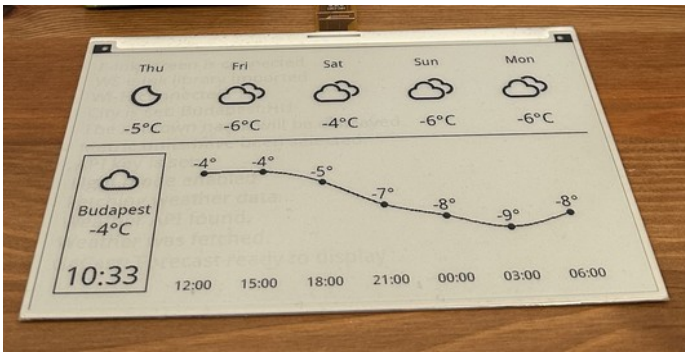


6. Wait 30sec

The **Status Screen** will now **appear** on the display, showing whether everything has been set up correctly.
If something is wrong, the status message will be very helpful in identifying the issue.
If everything is set up correctly, you should see the following:



After that, the weather display screen will appear.



If you see the weather data screen, everything is working correctly and you can begin assembling the hardware by following the other PDF tutorial.

If something is not working, you can contact me for help through the following channels.
(Please include a photo of your status screen to help speed up the troubleshooting process.)

MakerWorld:
https://makerworld.com/en/@EBengineering

Project page:
https://makerworld.com/en/crowdfunding/107-g4cast-e-ink-weather-station

I hope you've been enjoying the build process so far! The most challenging part is now behind you, and the hardest steps are complete. From here on, assembling your G4Cast Weather Station should be smooth and straightforward. You're well on your way to seeing your weather data come to life, and I'm confident you'll have your fully functional station up and running in no time. Keep going—you're almost there!



Best Regards,
EBdesign